

Description of the PyCh architecture for the Autonomous Vehicle Storage and Retrieval System (AVS/RS) model

Model Architecture Overview

The architecture is organized hierarchically into three main operational layers: the Global System Layer, the Aisle Layer, and the Tier Layer. Communication across these layers relies on dedicated physical and logical channels that coordinate the physical movement of inventory units (totes) and synchronize processes, including conveyors, lifts, and vehicles.

Global Layer and Demand Generation

At the top level, the system workload is driven by two independent components: the Storage Generator and the Retrieval Generator. These processes create incoming orders following an exponential inter-arrival time distribution. Totes are initialized with unique identifiers and specific metadata regarding their operations.

The Storage Generator passes items via the `c_store_order` channel to the Storage Controller, which assigns specific storage coordinates (aisle, tier, column, side) based on a configured storage policy. Similarly, the Retrieval Generator communicates via the `c_retrieve_order` channel with the Retrieval Controller, which selects an occupied rack location according to a retrieval policy. Both controllers maintain internal waitlists and track tier buffer capacities using the logical feedback channels `c_inbound_changed` and `c_outbound_changed` to safely manage flow control and avoid deadlocks.

Aisle Layer and Vertical Transport

Once an order is routed to a specific aisle, it transitions into the Aisle Layer. For storage operations, the Storage Controller dispatches the tote via `c_store_to_conv`, which feeds into the storage Conveyor process. This conveyor introduces a physical transit timeout before passing the tote to the Lift process via the `c_conv_to_lift` channel. The Lift handles vertical transitions; it collects the storage tote from the baseline level and moves it to the designated tier, depositing it into the Tier Buffer via `c_lift_to_buf`.

For retrieval operations, the Lift communicates with the Tier Buffer using the logical synchronization channels `c_lift_ready` and `c_next_tier`. After receiving a target tier notification, the Lift retrieves the outbound tote from the buffer via `c_buf_to_lift`, brings it down to the ground level, and transfers it to the retrieval Conveyor via `c_lift_to_conv`. This conveyor eventually routes the completed item out of the aisle system via `c_retrieve_done`.

Tier Layer and Horizontal Transport

The Tier Layer represents the finest granularity of the model, where horizontal movements take place within individual levels of an aisle. The Buffer process acts as the core intersection here, managing inbound and outbound queues up to a predefined capacity constraint.

For storage, the Vehicle process requests a tote from the buffer using the `c_selected_tote` channel, extracts it from the buffer via `c_buf_to_veh`, and simulates horizontal travel along the aisle to place it into the rack. Upon completion, the vehicle triggers the `c_store_done` channel. For retrievals, orders from the Retrieval Controller arrive at the Demand Buffer via `c_retrieve_to_dem`. The Demand Buffer coordinates these incoming requests with status

updates from the main tier buffer via `c_buf_to_dem` and dispatches commands to the Vehicle via `c_dem_to_veh`. The Vehicle then drives to the target rack column, picks up the tote, returns to the tier buffer, and deposits it using `c_veh_to_buf`.

System Performance Monitoring

The final phase of the modeling architecture consists of statistical collection handled by the Storage Monitor and the Retrieve Monitor processes. The Storage Monitor listens to the global `c_store_done` channel to calculate the total flow time of storage operations. The Retrieve Monitor listens to the array of `c_retrieve_done` channels coming from all active aisles to measure retrieval performance. Both monitors dynamically calculate the average and standard deviation of flow times while tracking system throughput. The simulation continues running until the standard deviation of the throughput over a set number of minimum samples falls below a predefined stability threshold, signaling that the system has reached a steady state.